

Production Residual Family Head Warm-Start from Deployed Family-Head Model

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Claim: converting the deployed family-aware Stage-3 model into a *residual family head*

$$\mu(x, f) = \mu_{\text{shared}}(h(x)) + \Delta_f(h(x))$$

improves the production point-table metrics while preserving the deployment path for new injector families.

Best run: `delta_L2 = 1e-4`.

Primary metric: CDF uncensored RMSE improves 5.056 → 4.631 mm.

Secondary metrics: P50 RMSE improves 3.931 → 3.395 mm; CDF ECE improves 0.0708 → 0.0604.

Output:

MLP/runs_mlp/residual_family_head_production_20260530_225256/

Thesis/slides/slides_residual_family_head_production/

Experiment Contract

Fixed production baseline

- Baseline run: `distill_cdf_family_head_v3_FROZEN_anchor_off_20260529_213346`
- Feature contract: `a_dp050_plus_pressures`
- Evaluation: canonical point tables (`cdf_uncensored`, `p50_observed`, `q1_grid_all`)
- Device: CUDA, seed inherited from the production run

Ablation	Values
Residual shrinkage	0, $1e-4$, $1e-3$, $1e-2$
Mimic warm-up	10 epochs, teacher is the deployed production model
Refinement	150 max epochs, patience 20, trunk frozen
Trainable params	49,923 params: residual deltas + shared log-var head

Architecture: Residual Family Head

Current family head

$$\mu(x, f) = \mu_f(h(x))$$

Each family owns the full mean decoder. A new family needs a fallback policy or a trained head.

Residual family head

$$\mu(x, f) = \mu_{\text{shared}}(h(x)) + \Delta_f(h(x))$$

The shared mean is copied from the modified-family production head (`mu_heads.1`); each residual head is zero-initialised.

Warm-start mapping

- `copy trunk.*`
- `copy log_var_head.*`
- `mu_heads.1` → `mu_shared_head`
- `delta_heads.*`: zero init, then KD mimic

Deployment property

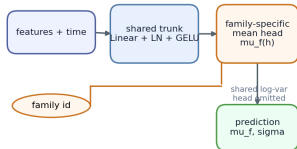
An unseen family with zero residual naturally predicts μ_{shared} ; no hard-coded `fallback_family_id` is needed.

Architecture: Family Head vs Residual vs FiLM

MLP family-conditioning architectures for spray penetration post-training

A. Family Head MLP

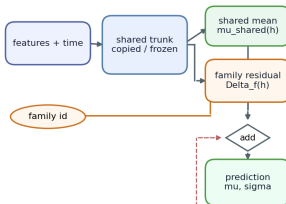
Family id selects the full mean decoder



Unseen family needs a fallback policy or a newly trained head.

B. Residual Family Head

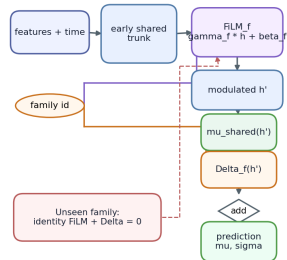
Shared mean plus a small family residual



Unseen family: $\Delta_f = 0$ fallback

C. Residual-FiLM

Family also modulates the hidden representation



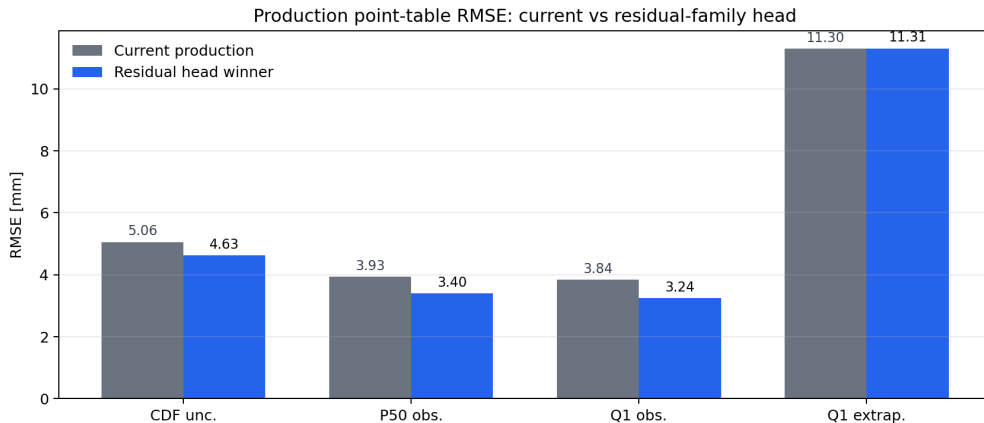
Unseen family: identity FiLM + $\Delta_f = 0$

Training Log Read

delta L2	mimic epochs	stop epoch	test loss	test delta RMS
0	10	104	2.074	0.816
1e-4	10	44	2.085	0.818
1e-3	10	110	2.070	0.813
1e-2	10	84	2.083	0.810

- All four runs completed and early-stopped cleanly.
- The mimic stage reduced the initial production-head mismatch before raw refinement.
- Delta magnitudes stayed tightly clustered ($\text{RMS} \approx 0.81$ in A-scaled space), so the shrinkage sweep mainly changes calibration and tail behaviour, not gross residual size.

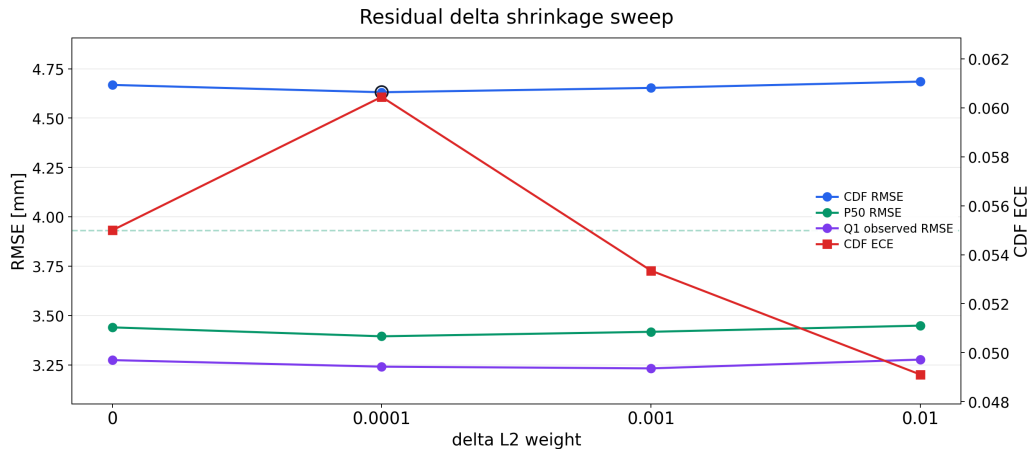
Headline Metrics: Current vs Winner



Headline Metrics: Current vs Winner: Reading

Model	CDF RMSE	CDF ECE	CDF CRPS	P50 RMSE	Q1 obs RMSE
Current production	5.056	0.0708	2.801	3.931	3.839
Residual winner (1e-4)	4.631	0.0604	2.382	3.395	3.242

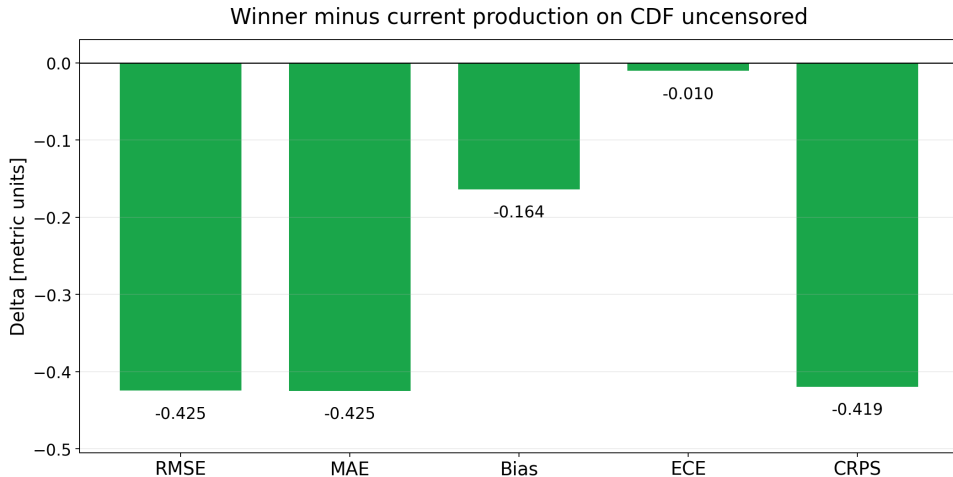
Shrinkage Sweep



Shrinkage Sweep: Reading

- All residual settings beat current production on CDF RMSE.
- $1e-4$ gives the best primary RMSE: 4.631 mm.
- Stronger shrinkage improves CDF ECE and Q1 extrapolated RMSE, but gives back a small amount of primary CDF RMSE.

CDF Uncensored Delta vs Production



CDF Uncensored Delta vs Production: Reading

Reading

- Winner reduces CDF RMSE by 0.425 mm and MAE by 0.425 mm.
- Bias moves closer to zero: $+0.343 \rightarrow +0.180$ mm.
- Calibration also improves: ECE $0.0708 \rightarrow 0.0604$, CRPS $2.801 \rightarrow 2.382$.

- 1 The deployed production model can be hot-started into a residual family head without retraining the trunk.
- 2 The best residual model beats current production on the primary metric: CDF uncensored RMSE $5.056 \rightarrow 4.631$ mm.
- 3 P50 and observed-window Q1 also improve substantially: P50 $3.931 \rightarrow 3.395$ mm; Q1 observed $3.839 \rightarrow 3.242$ mm.
- 4 Extrapolated Q1 is effectively flat for the winner ($11.295 \rightarrow 11.307$ mm); stronger shrinkage improves this tail slightly at a small CDF-RMSE cost.
- 5 Next step: package the residual winner as the new candidate deployed model, then run a seed check or a small new-family calibration simulation.

Artifacts: residual_from_prod_eval_summary.csv; verdict.json; figs/production_rmse_comparison.png; figs/delta_l2_sweep.png; figs/residual_svgp_context_rmse_comparison.png; figs/residual_svgp_vs_current_best_deltas.png.

Addendum: Residual SVGP Follow-Up

What changed

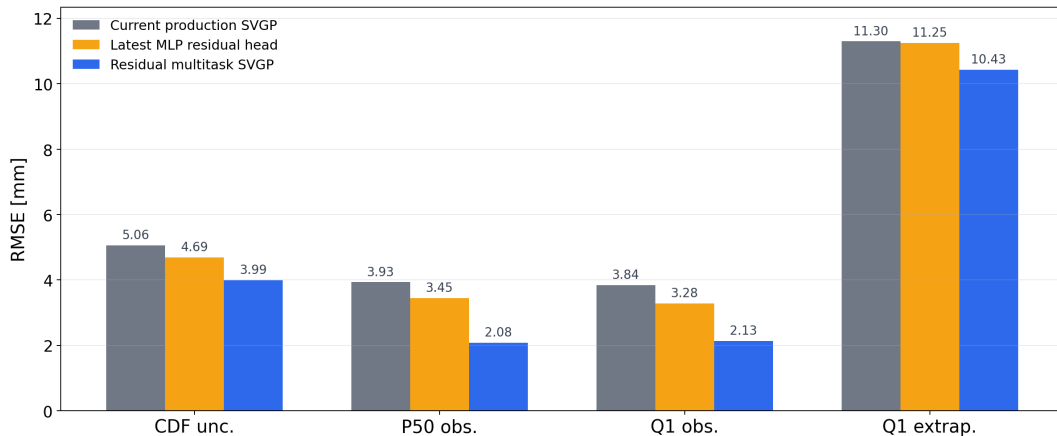
- Hot-started an additive residual multi-task SVGP from the deployed Stage-3 production checkpoint.
- Family id stays out of the kernel feature vector; it only routes the per-family residual GP.
- The winner is $\text{delta_L2} = 1\text{e-}4$; unseen families naturally fall back to the shared SVGP.

Model	delta L2	CDF RMSE	CDF ECE	CDF CRPS	P50 RMSE	Q1 extrap RMSE
MLP residual FH	1e-2	4.685	0.0491	2.368	3.449	11.248
Residual SVGP	1e-4	3.987	0.0262	2.040	2.077	10.429

Relative to the original SVGP baseline, the residual multitask winner also improves CDF uncensored RMSE from 4.193 mm to 3.987 mm.

Residual SVGP vs Current Best

RMSE context: current production SVGP vs current best MLP vs residual SVGP



Residual SVGP Sweep

delta L2	CDF RMSE	CDF ECE	CDF CRPS	P50 RMSE	Q1 extrap RMSE
0	4.193	—	—	2.649	10.591
1e-4	3.987	0.0262	2.040	2.077	10.429
1e-3	4.011	0.0300	2.061	2.095	10.474
1e-2	4.038	0.0347	2.093	2.108	10.513

- The sweep is narrow but stable: lower shrinkage gives the best CDF RMSE, while stronger shrinkage nudges calibration and tail extrapolation.
- The selected checkpoint sits at the sweet spot for the current deployment question: best primary RMSE and best calibration on the evaluated tables.

Next MLP-Only Ablation

Recommended architecture: residual FiLM adapter

Move the family signal from only the output head into the hidden representation while preserving exact hot-start compatibility:

$$h'_l = \gamma_{g,l} \odot h_l + \beta_{g,l}, \quad \mu(x, g) = \mu_{\text{shared}}(h'_L) + \Delta_g(h'_L)$$

- Identity init: $\gamma_{g,l} = 1$, $\beta_{g,l} = 0$, and $\Delta_g = 0$, so the copied production model is unchanged at step 0.
- Unknown family fallback: identity adapter plus zero residual gives the shared predictor without a hard-coded fallback family.
- MLP-specific question: does injector family change the representation of `a_dp050_plus_pressures`, not just the final mean offset?

MLP-Only Ablation Matrix

Mode	Placement	Trainable parts	Purpose
last-block FiLM	final trunk block	FiLM + delta heads + log-var	smallest representation-level test
all-block FiLM	every trunk block	FiLM + delta heads + log-var	pressure interactions by family
low-rank adapter	final/all blocks	rank-4/8 adapter + heads	more expressive few-shot adapter
log-var residual	uncertainty head	log-var residual only	calibration without mean-capacity change

Start with `residual_film_last_block`; target is to beat the latest MLP residual family-head CDF RMSE of 4.685 mm while keeping CDF ECE near 0.049. The detailed run matrix is stored in `mlp_only_architecture_ablation_candidates.csv`.

Residual-FiLM Production Result

Completed production sweep

The planned MLP-only ablation was run on the production dataset, without LONO. The source was the latest residual family-head checkpoint ($\text{delta_L2} = 1\text{e-}2$); the inserted FiLM adapter was identity-initialized and placed after the final trunk block.

Model	film L2	delta L2	CDF RMSE	CDF ECE	CDF CRPS	P50 RMSE	Q1 obs RMSE
Source residual FH	–	1e-2	4.685	0.0491	2.368	3.449	3.277
Residual-FiLM	1e-4	1e-3	4.510	0.0404	2.242	3.166	3.110

- CDF RMSE improves by **0.176 mm**; P50 improves by **0.283 mm**.
- Calibration improves as well: ECE $0.0491 \rightarrow 0.0404$, CRPS $2.368 \rightarrow 2.242$.
- Q1 extrapolated is effectively flat: $11.248 \rightarrow 11.265$ mm.

Residual-FiLM Sweep

film L2	delta L2	CDF RMSE	CDF ECE	P50 RMSE	Q1 extrap RMSE
1e-4	1e-4	4.562	0.0367	3.247	11.274
1e-4	1e-3	4.510	0.0404	3.166	11.265
1e-4	1e-2	4.554	0.0422	3.216	11.237
1e-3	1e-4	4.523	0.0410	3.183	11.238
1e-3	1e-3	4.574	0.0386	3.237	11.272
1e-3	1e-2	4.547	0.0426	3.209	11.243
1e-2	1e-4	4.525	0.0415	3.184	11.249
1e-2	1e-3	4.523	0.0427	3.174	11.216
1e-2	1e-2	4.523	0.0410	3.176	11.245

- All nine residual-FiLM settings beat the source residual family head on CDF RMSE.
- The best point-accuracy setting is lightly regularized FiLM plus medium delta shrinkage.
- Stronger FiLM shrinkage is competitive and slightly helps Q1 extrapolation, but gives back CDF RMSE.

LONO Validation: Residual Deployment Family

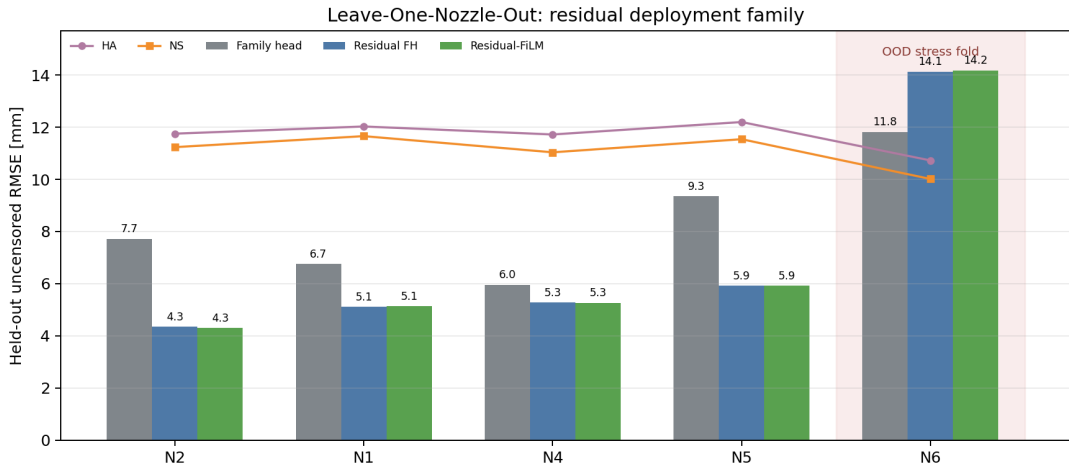
Protocol

- Hold out Nozzle1, 2, 4, 5, 6 one at a time; Nozzle0 stays in train as the sole first-generation injector-family anchor.
- Each fold trains Stage-1 $\rightarrow$ Stage-2 $\rightarrow$ Stage-3 from scratch with the held-out nozzle removed; no production-checkpoint leakage.
- Arms: family-head teacher, residual FH ($\text{delta_L2}=1\text{e-}4$), residual-FiLM ($\text{film_L2}=1\text{e-}4$, $\text{delta_L2}=1\text{e-}3$).

Aggregation	Family head	Residual FH	Residual-FiLM	HA / NS
All 5 folds	8.32 ± 2.33	6.95 ± 4.05	6.96 ± 4.08	11.69 / 11.10
4 folds excl. N6	7.44 ± 1.46	5.16 ± 0.65	5.15 ± 0.67	11.93 / 11.37

The residual win survives LONO on every in-family fold; N6 is reported as an OOD stress fold rather than hidden inside the mean.

LONO Validation: Per-Fold Read



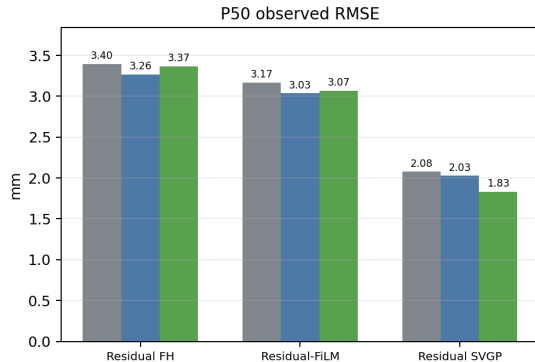
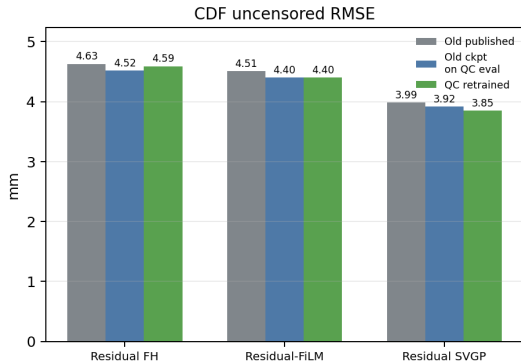
LONO Validation: Verdict

- On in-family held-out nozzles, residual FH improves family-head RMSE on every fold: N2 **-3.37 mm**, N1 **-1.64 mm**, N4 **-0.68 mm**, N5 **-3.43 mm**.
- The residual conversion also repairs the family-head over-prediction bias: the 4-fold mean bias moves from **+5.07 mm** to **+0.78 mm**.
- Residual FH and residual-FiLM are statistically indistinguishable under LONO. FiLM's in-distribution edge does not translate into unseen-nozzle transfer.
- N6 is a true stress fold: all MLP arms fail against HA/NS, and the residual arms under-predict severely. This motivates a separate OOD/nozzle-family study rather than another adapter sweep.

Artifacts: `lono_modified_only/lono_modified_only_summary.md`;
`lono_modified_only/lono_modified_only_comparison.csv`.

Dataset QC-Gate Ablation

Production Dataset QC-Gate Ablation



Dataset QC-Gate Ablation: Reading

Model	Old CDF	Old ckpt on QC	QC retrain CDF	Old P50	Old ckpt on QC	QC retrain P50
Residual FH	4.631	4.519	4.590	3.395	3.264	3.366
Residual-FiLM	4.510	4.401	4.400	3.166	3.035	3.065
Residual SVGP	3.987	3.921	3.848	2.077	2.026	1.829

- The QC-gated point tables improve all old checkpoints, so part of the gain is target cleanup rather than retraining.
- MLP retraining on the QC-gated root is neutral: residual FH regresses relative to the old checkpoint on the same QC table; residual-FiLM is effectively tied.
- Residual SVGP gains beyond target cleanup: CDF RMSE 3.921 $\rightarrow$ 3.848 mm and P50 RMSE 2.026 $\rightarrow$ 1.829 mm on the QC table.

Artifact: dataset_ablation_qc_gated/dataset_qc_gate_production_comparison.csv.